

Sharp Physical Repelling-Return Affine-Leg Remainder

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Abstract

The first physical endpoint leg of RH-324 produces an actual finite-noise prefix near the repelling point. We pass that same prefix through the second physical row and compare only this row with the RH-323 repelling tangent row, retaining the incoming coordinate. After including the Jacobian, the scaled physical row has an exact two-lobe formula. Its L^1 distance to the full Gaussian at the curved image is exactly two state-boundary tails, while removal of the quadratic displacement costs $4\Phi(u_c\sigma u^2/2) - 2$. We prove a uniform fourth moment for the actual first-leg marginal; this is the missing uniform-integrability input that ordinary L^1 convergence does not provide. For either repelling orientation,

$$\frac{D_{\sigma, a_\sigma}^\pm}{\sigma} \longrightarrow \sqrt{\frac{2}{\pi}} u_c \int_{\{\pm u > 0\}} u^2 p_d(u) du > 0, \quad a_\sigma \longrightarrow d.$$

The sum has a closed coefficient. Thus exponential, or even $o(\sigma)$, accuracy is false for this exact second hybrid Duhamel row term. The coefficient vanishes at the single row $u = 0$, and a source moving to the critical partition disproves a global uniform $O(\sigma)$ row estimate. No fully physical-to-fully affine two-leg equality, all-cycle transport, cyclic trace estimate, or Gate A–E promotion is asserted.

1 Physical row and repelling coordinates

Let u_c be the root in $(1, 2)$ of

$$u^3 - 2u^2 + 2u - 2 = 0,$$

and put

$$f(x) = 1 - u_c x^2, \quad r = u_c - 1, \quad \lambda = 2u_c r, \quad \alpha = 2u_c, \quad b = u_c^{-1/2}. \quad (1)$$

Numerically,

$$\begin{aligned} u_c &= 1.543689012692076\dots, & r &= 0.543689012692076\dots, \\ \lambda &= 1.678573510428322\dots, & b &= 0.804859535112210\dots \end{aligned}$$

Write ϕ , Φ , and $\bar{\Phi} = 1 - \Phi$ for the standard normal density, distribution, and survival functions. The archived row-normalized folded kernel on $[0, 1]$ is

$$P_\sigma(x, y) = \frac{\phi_\sigma(y - f(x)) + \phi_\sigma(y + f(x))}{Z_\sigma(x)}, \quad \phi_\sigma(t) = \sigma^{-1} \phi(t/\sigma), \quad (2)$$

where

$$Z_\sigma(x) = 1 - \bar{\Phi}\left(\frac{1 - f(x)}{\sigma}\right) - \bar{\Phi}\left(\frac{1 + f(x)}{\sigma}\right). \quad (3)$$

This is the probability that the folded Gaussian lies in the physical state interval; in particular $0 < Z_\sigma(x) \leq 1$.

Use repelling coordinates

$$x = r + \sigma u, \quad y = r + \sigma w, \quad I_\sigma = \left[-\frac{r}{\sigma}, \frac{1-r}{\sigma} \right]. \quad (4)$$

The quadratic map gives the exact, rather than asymptotic, expansion

$$f(r + \sigma u) = r - \lambda \sigma u - u_c \sigma^2 u^2 = r - \sigma d_\sigma(u), \quad d_\sigma(u) = \lambda u + u_c \sigma u^2. \quad (5)$$

The Jacobian in the next definition is essential.

Definition 1.1 (Second physical, curved, and tangent rows). For $u \in I_\sigma$, define the scaled physical row

$$\begin{aligned} L_\sigma(u, w) &:= \sigma P_\sigma(r + \sigma u, r + \sigma w) \\ &= \frac{\phi(w + d_\sigma(u)) + \phi(w + \frac{2r}{\sigma} - d_\sigma(u))}{Z_\sigma(r + \sigma u)} \mathbf{1}_{I_\sigma}(w), \end{aligned} \quad (6)$$

and, on the full line,

$$C_{\sigma,u}(w) = \phi(w + d_\sigma(u)), \quad A_u(w) = \phi(w + \lambda u). \quad (7)$$

All three rows have mass one.

The first lobe in (6) is centered at the signed exact image; that image is positive in the fixed local repelling chart near r . The other lobe is the folded companion. No symmetry with the endpoint expansion of RH-324 has been assumed.

2 Exact row identities and their sharp local limit

Theorem 2.1 (Exact repelling-row identities). *For every $\sigma > 0$ and every physical source $u \in I_\sigma$,*

$$\|L_\sigma(u, \cdot) - C_{\sigma,u}\|_1 = 2 \left[\bar{\Phi}\left(\frac{r}{\sigma} - d_\sigma(u)\right) + \bar{\Phi}\left(\frac{1-r}{\sigma} + d_\sigma(u)\right) \right]. \quad (8)$$

Moreover,

$$\|C_{\sigma,u} - A_u\|_1 = 4\Phi\left(\frac{u_c \sigma u^2}{2}\right) - 2. \quad (9)$$

Consequently, if $B_\sigma(u)$ and $S_\sigma(u)$ denote the right sides of (8) and (9), then

$$\max\{0, S_\sigma(u) - B_\sigma(u)\} \leq \|L_\sigma(u, \cdot) - A_u\|_1 \leq S_\sigma(u) + B_\sigma(u). \quad (10)$$

Proof. The two unnormalized lobes in (2) have total mass one on $[0, \infty)$, so restriction to $[0, 1]$ gives $Z_\sigma \leq 1$. On I_σ , the numerator of L_σ contains $C_{\sigma,u}$ and division by $Z_\sigma \leq 1$ only increases it. Hence $L_\sigma \geq C_{\sigma,u}$ on I_σ , while $L_\sigma = 0$ off that interval. Since both are probability densities, the positive and negative parts of their difference have equal mass. Each mass is the amount of $C_{\sigma,u}$ outside I_σ , namely

$$\bar{\Phi}(r/\sigma - d_\sigma(u)) + \bar{\Phi}((1-r)/\sigma + d_\sigma(u)).$$

This proves (8); in particular the companion lobe, state restriction, and exact normalizer have already been combined.

The centers in (7) differ by $u_c \sigma u^2$. Unit Gaussians separated by t have unhalved L^1 distance $4\Phi(|t|/2) - 2$, which proves (9). The ordinary and reverse triangle inequalities give (10). $\square$

Corollary 2.2 (Sharp fixed-row coefficient). *For each fixed $u \in \mathbb{R}$,*

$$\lim_{\sigma \downarrow 0} \frac{\|L_\sigma(u, \cdot) - A_u\|_1}{\sigma} = \sqrt{\frac{2}{\pi}} u_c u^2. \quad (11)$$

At $u = 0$ the coefficient is zero; for every fixed $u \neq 0$ it is positive.

Proof. For fixed u , both arguments of the survival functions in (8) are a positive constant times σ^{-1} up to $O(1)$. Thus $B_\sigma(u) = o(\sigma)$. Taylor expansion at zero gives

$$\frac{S_\sigma(u)}{\sigma} \longrightarrow 2\phi(0)u_c u^2 = \sqrt{2/\pi} u_c u^2.$$

Now use (10). □

The fixed-source quantifier cannot be made global.

Proposition 2.3 (No global uniform $O(\sigma)$ row theorem). *Let*

$$u_\sigma = \frac{b - r}{\sigma},$$

so that the physical source $r + \sigma u_\sigma$ equals the critical partition b . Then

$$\liminf_{\sigma \downarrow 0} \|L_\sigma(u_\sigma, \cdot) - A_{u_\sigma}\|_1 \geq 1. \quad (12)$$

In particular, $\sup_{u \in I_\sigma} \|L_\sigma(u, \cdot) - A_u\|_1$ is not $O(\sigma)$.

Proof. Since $f(b) = 0$, (5) gives $d_\sigma(u_\sigma) = r/\sigma$. Therefore $B_\sigma(u_\sigma) \rightarrow 1$: the curved Gaussian has half its mass beyond the state boundary, while the other boundary tail vanishes. On the other hand, the curved-to-tangent displacement is

$$u_c \sigma u_\sigma^2 = \frac{u_c (b - r)^2}{\sigma} \longrightarrow \infty,$$

so $S_\sigma(u_\sigma) \rightarrow 2$. The lower bound in (10) yields (12). □

3 The actual physical first-leg prefix

We now specify the incoming law rather than reusing a local seed. Put $L = \sigma^{-1}$ and, for $0 \leq a \leq L$, let the exact RH-322 entrance density be

$$\tilde{h}_{\sigma,a}(v) = \frac{\phi(v - a) + \phi(2L - v - a)}{\Phi(a) - \bar{\Phi}(2L - a)} \mathbf{1}_{[0,L]}(v). \quad (13)$$

For $c_\sigma(v) = \alpha v - u_c \sigma v^2$, the exact RH-324 first physical row is

$$K_\sigma(v, u) = \frac{\phi(u + 2r/\sigma - c_\sigma(v)) + \phi(u + c_\sigma(v))}{Z_\sigma(1 - \sigma v)} \mathbf{1}_{I_\sigma}(u). \quad (14)$$

Define its actual U -marginal by

$$\mu_{\sigma,a}(du) = \left\{ \int_0^L \tilde{h}_{\sigma,a}(v) K_\sigma(v, u) dv \right\} du. \quad (15)$$

This is an exact physical prefix, not the affine U -law.

For $d \geq 0$, the RH-323 affine entrance and intermediate densities are

$$g_d(v) = \frac{\phi(v-d)}{\Phi(d)} \mathbf{1}_{[0,\infty)}(v), \quad (16)$$

$$p_d(u) = \frac{1}{s_1 \Phi(d)} \phi\left(\frac{u+\alpha d}{s_1}\right) \Phi\left(\frac{d-\alpha u}{s_1}\right), \quad s_1 = \sqrt{1+\alpha^2}. \quad (17)$$

Equivalently, p_d is the law of

$$U = -\alpha V - Z_1, \quad V \sim g_d, \quad Z_1 \sim N(0, 1), \quad (18)$$

with V and Z_1 independent. RH-324 proves, for fixed $0 < \rho < 1 - b$ and

$$m_\rho = u_c(1-\rho)^2 - 1 > 0, \quad M_2(a) = a^2 + 1 + a \frac{\phi(a)}{\Phi(a)},$$

the explicit marginal consequence

$$\begin{aligned} \|\mu_{\sigma,a} - p_d\|_1 &\leq |a-d| + \frac{2\bar{\Phi}(L-a)}{\Phi(a)} + \sqrt{\frac{2}{\pi}} u_c \sigma M_2(a) \\ &\quad + 2\{\bar{\Phi}(m_\rho/\sigma) + \bar{\Phi}((1-r)/\sigma)\} + \frac{2\bar{\Phi}(\rho/\sigma - a)}{\Phi(a)}. \end{aligned} \quad (19)$$

Thus $a_\sigma \rightarrow d$ implies ordinary L^1 convergence. The next lemma supplies the weighted convergence that (19) alone does not imply.

Lemma 3.1 (Uniform fourth moment of the actual prefix). *For every compact $\mathcal{K} \subset [0, \infty)$ there are $\sigma_0, C_{\mathcal{K}} > 0$ such that*

$$\sup_{0 < \sigma < \sigma_0} \sup_{a \in \mathcal{K}} \int_{I_\sigma} |u|^4 \mu_{\sigma,a}(du) \leq C_{\mathcal{K}}. \quad (20)$$

Consequently, whenever $a_\sigma \rightarrow d$,

$$\int_{\{\pm u > 0\}} u^2 \mu_{\sigma,a_\sigma}(du) \longrightarrow \int_{\{\pm u > 0\}} u^2 p_d(u) du. \quad (21)$$

Proof. Fix $0 < \rho < 1 - b$ and put

$$q_0 = \max\{r, 1-r\}, \quad z_0 = \Phi(1) - \frac{1}{2} > 0.$$

The explicit density (13) gives, uniformly for $a \in \mathcal{K}$ and small σ ,

$$\mathbb{E}_{\tilde{h}_{\sigma,a}} V^8 \leq C_{\mathcal{K}}. \quad (22)$$

Indeed, its denominator is bounded below, the first numerator term is a Gaussian with bounded center, and the reflected term begins at $L - \sup \mathcal{K}$ after reflection and is Gaussian-tail small.

For every $x \in [0, 1]$ and small σ , the folded normalizer is at least z_0 : the corresponding standard-normal interval has length at least two and contains zero. On $v \leq \rho/\sigma$, the source $1 - \sigma v$ remains in the negative endpoint branch and its image has modulus at least m_ρ . Integrating the two numerator lobes in (14) therefore gives

$$\begin{aligned} \mathbb{E}(|U|^4 \mid V = v) &\leq z_0^{-1} \{c_\sigma(v)^4 + 6c_\sigma(v)^2 + 3 \\ &\quad + q_0^4 \sigma^{-4} \bar{\Phi}(m_\rho/\sigma)\}. \end{aligned} \quad (23)$$

Here the first three terms are the fourth moment of the full main Gaussian; the last term bounds the remote folded lobe on the physical support. Moreover,

$$|c_\sigma(v)|^4 \leq 8\alpha^4 v^4 + 8u_c^4 \sigma^4 v^8. \quad (24)$$

Equations (22)–(24) make the contribution of $v \leq \rho/\sigma$ uniformly bounded.

On the complementary entrance tail, $|U| \leq q_0/\sigma$ and Markov's inequality applied to (22) gives

$$\Pr\{V > \rho/\sigma\} \leq \rho^{-8} \sigma^8 \mathbb{E}V^8.$$

Its contribution to $\mathbb{E}|U|^4$ is therefore $O(\sigma^4)$. This proves (20).

For the second assertion, truncate $u^2 \mathbf{1}_{\{\pm u > 0\}}$ at $|u| \leq M$. The error between the truncated μ_{σ, a_σ} and p_d integrals is at most $M^2 \|\mu_{\sigma, a_\sigma} - p_d\|_1$. The two omitted tails are at most M^{-2} times their fourth moments; p_d has all moments by (18). First let $\sigma \downarrow 0$ using (19), then let $M \rightarrow \infty$. This proves (21) separately for both signs. $\square$

4 Sharp transported second-hybrid row term

The data type of the main result is fixed before measuring its error. The two retained (u, w) path laws

$$\mathbf{H}_{\sigma, a}^{\text{phys}}(du, dw) = \mu_{\sigma, a}(du) L_\sigma(u, w) dw, \quad \mathbf{H}_{\sigma, a}^{\text{tan}}(du, dw) = \mu_{\sigma, a}(du) A_u(w) dw \quad (25)$$

have the *same actual physical first-leg prefix*. Only the second row changes. This is precisely the second hybrid row term in the retained-path Duhamel telescope of RH-325.

For the two repelling orientations define

$$D_{\sigma, a}^\pm = \int_{\{\pm u > 0\}} \|L_\sigma(u, \cdot) - A_u\|_1 \mu_{\sigma, a}(du), \quad D_{\sigma, a} = D_{\sigma, a}^+ + D_{\sigma, a}^-. \quad (26)$$

Because u is retained, $D_{\sigma, a}^\pm$ is exactly the unhalved L^1 norm of the difference in (25) restricted to the corresponding sector.

Lemma 4.1 (The physical boundary is negligible under the actual prefix). *Uniformly for a in a fixed compact subset of $[0, \infty)$,*

$$\int B_\sigma(u) \mu_{\sigma, a}(du) = o(\sigma). \quad (27)$$

Proof. Choose $0 < \varepsilon < \min\{r, b - r\}$ and set

$$m_\varepsilon = \min\{f(r + \varepsilon), 1 - f(r - \varepsilon)\} > 0.$$

When $|u| \leq \varepsilon/\sigma$, the two survival-function arguments in (8) are exactly $f(r + \sigma u)/\sigma$ and $(1 - f(r + \sigma u))/\sigma$. Hence $B_\sigma(u) \leq 4\bar{\Phi}(m_\varepsilon/\sigma)$ there. Off this local chart, $B_\sigma \leq 2$, while lemma 3.1 gives

$$\mu_{\sigma, a}\{|u| > \varepsilon/\sigma\} \leq C_K \sigma^4 / \varepsilon^4.$$

Thus the integral is at most $4\bar{\Phi}(m_\varepsilon/\sigma) + 2C_K \sigma^4 / \varepsilon^4 = o(\sigma)$. $\square$

Theorem 4.2 (Sharp physical repelling-return hybrid remainder). *Let $\sigma \downarrow 0$ and let $a_\sigma \geq 0$ satisfy $a_\sigma \rightarrow d \geq 0$. Then, for each sign,*

$$\boxed{\frac{D_{\sigma, a_\sigma}^\pm}{\sigma} \longrightarrow \sqrt{\frac{2}{\pi}} u_c \int_{\{\pm u > 0\}} u^2 p_d(u) du > 0.} \quad (28)$$

Summing the orientations gives the closed formula

$$\boxed{\frac{D_{\sigma, a_\sigma}}{\sigma} \longrightarrow \sqrt{\frac{2}{\pi}} u_c \left[1 + \alpha^2 \left(d^2 + 1 + d \frac{\phi(d)}{\Phi(d)} \right) \right] > 0.} \quad (29)$$

Proof. Let $E_\sigma(u) = \|L_\sigma(u, \cdot) - A_u\|_1$ and let $S_\sigma(u)$ be (9). The reverse and ordinary triangle inequalities give $|E_\sigma(u) - S_\sigma(u)| \leq B_\sigma(u)$. Therefore lemma 4.1 shows, in either sector,

$$\left| \int_{\{\pm u > 0\}} (E_\sigma - S_\sigma) d\mu_{\sigma, a_\sigma} \right| = o(\sigma). \quad (30)$$

The elementary Gaussian-shift inequality

$$0 \leq 4\Phi(t/2) - 2 \leq \sqrt{\frac{2}{\pi}} t, \quad t \geq 0,$$

implies

$$0 \leq \frac{S_\sigma(u)}{\sigma} \leq \sqrt{\frac{2}{\pi}} u_c u^2, \quad \frac{S_\sigma(u)}{\sigma} \longrightarrow \sqrt{\frac{2}{\pi}} u_c u^2. \quad (31)$$

The convergence is uniform on each fixed bounded u -interval. On such an interval, (19) transfers the integral from μ_{σ, a_σ} to p_d . Outside it, (31) and the uniform fourth moment make the second-moment tails uniformly small. Thus

$$\int_{\{\pm u > 0\}} \frac{S_\sigma(u)}{\sigma} \mu_{\sigma, a_\sigma}(du) \longrightarrow \sqrt{\frac{2}{\pi}} u_c \int_{\{\pm u > 0\}} u^2 p_d(u) du.$$

Together with (30), this proves (28). The density (17) is strictly positive on $\mathbb{R}$, so both sector coefficients are strictly positive.

Finally (18) gives

$$\mathbb{E}U^2 = 1 + \alpha^2 \mathbb{E}V^2, \quad \mathbb{E}V^2 = d^2 + 1 + d \frac{\phi(d)}{\Phi(d)}.$$

Adding the two sector moments proves (29). □

Corollary 4.3 (Scoped negative and fixed first-alias phase). *For every $d \geq 0$ and along every sequence $a_\sigma \rightarrow d$, the exact second hybrid row error in (26) is neither exponentially small nor $o(\sigma)$. If the inherited first-alias clearance satisfies*

$$\delta_k = C_b \lambda^{-2k} \{1 + o(1)\}, \quad C_b > 0,$$

and

$$a_\sigma = \frac{\delta_k}{\sigma}, \quad \eta_\sigma := k - \frac{\log(1/\sigma)}{2 \log \lambda} \longrightarrow \eta,$$

then $a_\sigma \rightarrow d = C_b \lambda^{-2\eta}$ and (28)–(29) hold with this phase-matched d .

The comparison $\sigma = o(R^{-2k})$ for $R = 1.4$ and $k = \log(1/\sigma)/(2 \log \lambda) + O(1)$ remains valid because $\log R / \log \lambda = 0.6496301165 \dots < 1$. This is only local scale compatibility. The positive coefficient is an obstruction to improving the hybrid term below order σ , not an obstruction at the larger first-alias target scale.

5 Deterministic reproduction and claim boundary

The implementation evaluates the exact normalizer, both row identities, triangle bounds, the density p_d , and its two sector moments using standard-library composite Simpson quadrature. The displayed sector integrals use the finite numerical window $|u| \leq 32$; the exact total uses the closed formula in (29). Selected coefficients are

d	$u < 0$ coefficient	$u > 0$ coefficient	exact total
0	12.88903630	0.08295737	12.97199367
0.5	18.83879635	0.05712449	18.89592085
1	28.05382711	0.03498685	28.08881396

The sector sums agree with the closed total to the displayed digits. Finite curvature-proxy rows approach these values and exact row quadrature lies inside (10). These rows check formulas; no exponent is fitted and no finite table is used to prove an asymptotic statement.

The signs $U < 0$ and $U > 0$ label the two orientations on either side of the repelling coordinate r . They are *not* the two RH-19 critical siblings, which live in $\sqrt{\sigma}$ -scale windows around the distinct partition point b . The folded companion lobe in (6) is already included in the exact boundary identity and is not assigned a parity weight.

Remark 5.1 (Exact data-type firewall). The equality represented by $D_{\sigma,a}^{\pm}$ belongs only to the second hybrid Duhamel row term with the same actual first-leg prefix and retained u . Marginalizing u gives only the contraction bound

$$\|\mathbf{H}_{\sigma,a,W}^{\text{phys}} - \mathbf{H}_{\sigma,a,W}^{\text{tan}}\|_1 \leq D_{\sigma,a}.$$

Replacing the first physical leg as well creates a separate hybrid term; there is no asserted equality between the fully physical two-leg law and the fully affine two-leg law. Nor does the theorem supply a global uniform row bound, an all-cycle $O(k\sigma)$ estimate, phase transport through $2k$ legs, cyclic trace control, parity/shell cancellation, full-trace replacement, or determinant gluing.

Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt trace formula or completed-zeta divisor equality, and does not imply the Riemann Hypothesis. The next legitimate interface, RH-333, is to transport the common clearance phase and incoming moments through the full boundary cycle and either prove the retained-path $O(k\sigma)$ estimate on that one clock or isolate a genuine phase/stability obstruction. Its first admissible test is the possible escape gap in the raw mass-one forward all-affine chain. That test must be kept distinct from any cyclic bridge, Doob transform, or branch-complete non-Gaussian reference, which is not yet defined in the required physical data type.